

Norcem Brevik — Full-Scale CCS in the Cement Industry

Capturing Carbon from Norway's Oldest Cement Plant — A Blueprint for Industrial Decarbonisation

QUICK FACTS

- Sector:** Cement Manufacturing
- Location:** Brevik, Telemark, Norway
- Technology:** Amine post-combustion capture (Aker Carbon Capture); ship-based CO₂ transport to Northern Lights

1. Background: The Cement Problem

Cement is one of the most carbon-intensive materials on Earth. Global cement production emits approximately 2.5 billion tonnes of CO₂ per year — roughly 7–8% of total global GHG emissions. Unlike power plants where CO₂ comes entirely from fuel combustion, about 60% of cement's CO₂ comes from calcination — the unavoidable chemical decomposition of limestone ($\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$) when heated to produce clinker.

This means that no matter what fuel is used — coal, gas, biomass, or even 100% renewable electricity — cement production will continue to emit significant CO₂ as long as it uses limestone as a feedstock. CCS is, at present, the only technology capable of capturing these process emissions at scale.

Norcem's cement plant in Brevik, Telemark, is the oldest continuously operating cement producer in Norway, with production dating back to 1916. It produces approximately 1.2 million tonnes of clinker per year, with associated CO₂ emissions of around 850,000 tonnes annually.

Operator	Heidelberg Materials Norway (formerly Norcem)
Location	Brevik, Porsgrunn municipality, Telemark, Norway
Plant age	Operating since 1916
Annual clinker production	~1.2 million tonnes
Annual CO ₂ emissions	~850,000 tonnes/year
Capture capacity (Phase 1)	400,000 tonnes CO ₂ /year (full plant: ~800,000 t/year possible)
Capture technology	Aker Carbon Capture Just Catch™ amine process
CO ₂ destination	Northern Lights (ship transport → offshore geological storage)

Government support	Norwegian Longship programme — partial cost coverage
Project status	Full-scale capture commenced 2024

2. Technology: Amine Capture at a Cement Plant

The Brevik CCS installation uses the Aker Carbon Capture 'Just Catch™' technology, a proprietary variant of post-combustion amine scrubbing optimised for cement plant flue gases. Cement flue gas presents specific challenges compared to power plant flue gas:

Parameter	Power Plant Flue Gas	Cement Plant Flue Gas	Implication for CCS
CO ₂ concentration	10–15%	14–30%	Higher conc. = lower capture energy per tonne
Dust/particulates	Low (after ESP)	Higher; cement dust	Requires robust pre-treatment
SO ₂ content	Variable	Low (limestone absorbs SO ₂)	Less solvent degradation risk
Temperature	~120°C	~200–300°C raw; ~120°C treated	Cooling required before amine contact
Flue gas volume	Large (air diluted)	Smaller per tonne cement	Smaller absorber required per tonne CO ₂

The Aker Just Catch™ process uses a proprietary advanced amine solvent (ACCTM) that offers lower energy consumption than first-generation MEA solvents and greater resistance to cement flue gas contaminants. The system integrates with the cement plant's existing heat infrastructure, using waste heat from clinker cooling to partially offset the energy required for solvent regeneration — significantly improving the overall energy efficiency.

3. The CO₂ Transport Chain: Ship to Northern Lights

One of Brevik's most innovative aspects is its CO₂ transport solution. Unlike projects that rely on dedicated pipelines (often impractical for coastal industrial sites), Norcem Brevik ships its liquefied CO₂ to the Northern Lights receiving terminal at Oygarden, near Bergen.

- **Step 1 — Liquefaction:** The captured CO₂ (gas at ~1 bar, >99% purity) is cooled to approximately –52°C and compressed to ~6.5 bar, converting it to liquid CO₂. In liquid form, CO₂ occupies roughly 1/500th of its gas volume, making ship transport economic.
- **Step 2 — Intermediate storage:** Liquid CO₂ is stored in insulated onshore tanks at the Brevik facility. Tank storage allows continuous capture operation even when ships are not loading.
- **Step 3 — Ship loading:** A dedicated CO₂ carrier vessel moors at the Brevik jetty and loads liquid CO₂ from the tanks. The ship serves multiple customers (not just Norcem), improving vessel utilisation and reducing per-tonne transport costs.

- **Step 4 — Transport to Oygarden:** The CO₂ carrier transports liquid CO₂ approximately 250 km along the Norwegian coast to the Northern Lights receiving terminal.
- **Step 5 — Offshore injection:** At Oygarden, the CO₂ is transferred to onshore tanks, re-pressurised, and pumped via subsea pipeline to injection wells in the Johansen formation, approximately 2,600 m below the North Sea floor.

4. Economics and Financing

Norcem Brevik could not have been built without substantial public support. The Norwegian government's Longship programme funded approximately 80% of the project cost — acknowledging that the project's purpose was as much to demonstrate and de-risk the technology for global replication as to decarbonise Norcem's specific plant.

Cost Element	Estimated Range	Comment
Capture unit capital	€300–400 million	For 400,000 t/year first-phase capture
Liquefaction facility	€30–50 million	Required for ship-based transport
Annual operating cost	€30–50 million	Energy, chemicals, staff, maintenance
Transport cost (Northern Lights)	~€50–80/tonne	Northern Lights tariff for transport + storage
Norwegian government support	~80% of capital	Longship programme funding
Carbon revenue (EU ETS equivalent)	~€60–80/tonne	Norcem is EU ETS-covered facility

5. Lessons and Relevance for India

- **India's cement sector is massive:** India is the world's second-largest cement producer, with production exceeding 380 million tonnes per year. Indian cement accounts for approximately 6–7% of India's total CO₂ emissions — a sector that cannot be decarbonised without CCS.
- **Cost trajectory matters:** Norcem's current cost structure is heavily influenced by first-of-a-kind premium. Projections suggest that a similar CCS system on an Indian cement plant would cost \$60–90/tonne at scale — within reach of India's EU CBAM exposure after 2026.
- **CBAM is the key driver for India:** The EU's Carbon Border Adjustment Mechanism will make Indian cement, steel, and aluminium exports to the EU significantly more expensive from 2026 onwards unless they can demonstrate low-carbon production. CCS is a viable compliance pathway.
- **Shared infrastructure model:** India's coastal cement plants — particularly in Gujarat, Rajasthan, and Andhra Pradesh — could benefit from a shared CO₂ transport and storage infrastructure modelled on Northern Lights, if offshore geological storage is confirmed in the Arabian Sea or Bay of Bengal.

Key Takeaway: Norcem Brevik proves that CCS on cement is technically and commercially viable — with the right policy framework. For India's cement sector, the arrival of EU CBAM makes that policy pressure real.

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